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Reading & Writing

Module 1

27 Questions

Among saltwater fish species, there is a clear association between habitat latitude and morphological variety. While tropical species are _____ deep- bodied physical forms (body shapes that are laterally compressed but vertically extended), polar and temperate species are highly dispersed across the morphological spectrum.

1 Mark for Review

ABC

Which choice completes the text with the most logical and precise word or phrase?

A authenticated by

B concentrated among

C habituated to

D contemporary with

Although more documentaries are being made than ever before, too many seem _____, offering stories without any depth. In a refreshing contrast, Ahmir "Questlove" Thompson's 2021 documentary Summer of Soul (...or, When the Revolution Could Not Be Televised) draws from present-day interviews to supply compelling historical and social context for footage of the 1969 Harlem Cultural Festival.

2 Mark for Review

ABC

Which choice completes the text with the most logical and precise word or phrase?

A cryptic

B inaccessible

C haughty

D perfunctory

The following text is from Billie Jean King's 2021 autobiography Al In. [P]eople on both sides of my family had repeatedly demonstrated an independent streak. In the end, that was the temperament I gravitated toward, too. Both the Moffitts and the members of my mother's clan, the Jermans, came from mining and oil-geyser towns on the western frontier. They kept their heads down and worked, worked, worked. But they also bucked convention. ©2021 by Billie Jean King Enterprises, Inc.

3 Mark for Review ABC

As used in the text, what does the word "demonstrated" most nearly mean?

A Exhibited

B Defined

C Confirmed

D Protested

The following text is adapted from Edith Wharton's 1911 novella Ethan Frome. The narrator has asked the woman he rents a room from about Ethan Frome, a town resident he encountered recently. Her mind was a store-house of innocuous anecdote and any question about her acquaintances brought forth a volume of detail; but on the subject of Ethan Frome I found her unexpectedly reticent. There was no hint of disapproval in her reserve; I merely felt in her an insurmountable reluctance to speak of him.

4 Mark for Review

ABC

As used in the text, what does the word "reserve" most nearly mean?

A Misgiving

B Constraint

C Composure

D Modesty

Paul Linebarger is often recognized as an influential science fiction writer (under the pen name Cordwainer Smith), but his most significant work was a 1940s classified US Army guide exploring the role of propaganda in times of war. Citing the uniquely absorbing nature of motion pictures, he contended that US propaganda should equal popular films in its appeal, entertaining audiences while engendering views of the nation as an ally. Linebarger's work shaped the distinctive US approach to promoting national interests, one that continues to draw inspiration from elements of pop culture.

5 Mark for Review

ABC

Which choice best describes the overall structure of the text?

A. It distinguishes a work from others in its genre, outlines the central objective of that work, and then addresses a recent resurgence of interest in that work.

B. It sketches the development of a prominent work, familiarizes readers with an argument in that work, and then suggests that the work has had repercussions in other fields.

C. It characterizes a work as having been underappreciated, describes a core concept in that work, and then argues that the influence of that concept is recognizable in other works.

D. It introduces a significant work, summarizes a noteworthy idea in that work, and then places that idea into a context that demonstrates its broader implications.

On receiving good news, people's first instinct might be to immediately share it, but a recent experiment suggests that keeping it a secret can be psychologically beneficial. Study participants were randomly assigned to reflect on an experience in which they either shared good news or kept it to themselves. On average, those who kept good news secret reported feeling more energized by the experience than participants who shared did, perhaps, as the researchers suggest, because the choice to savor good news is usually a personal one and autonomous motivation is linked with feelings of vitality.

6 Mark for Review ABC

Which choice best describes the function of the underlined portion in the text as a whole?

A) It presents a scenario whose psychological impact was investigated in the study summarized in the text.

B) It indicates a reason why skepticism about the conclusions of the psychological study mentioned in the text is likely warranted.

C) It reports the main conclusion of a study into a social phenomenon whose methodology is critiqued in the text.

D) It provides an observation about human behavior that was validated by the experiment summarized in the text.

Alternative-history fiction is a subgenre of science fiction in which plots center on "what if?" questions. What if India had started the Industrial Revolution? What if Soviet cosmonauts had been first to land on the moon? Speculative counterfactuals like these can be great fodder for stories, but they're also commonly deployed by academic historians to better understand factors influencing historical events. Well-realized and coherent alternative-history stories can thus complement historians' speculations about the past.

7 Mark for Review

ABC

Which choice best describes the function of the underlined portion in the text as a whole?

A It provides examples of the types of scenarios addressed in alternative- history fiction.

B It illustrates how academic historians have influenced writers of alternative-history fiction.

C It indicates specific ideas that were first raised by writers of alternative- history fiction and that subsequently inspired scholarship by academic historians.

D It acknowledges counterfactual questions that academic historians have largely overlooked.

Text 1

In separate studies, Lingbo Meng and colleagues and Xinhua He and colleagues examined whether plants transfer nutrients to one another using a common mycorrhizal network (CMN)—a lattice of fungal strands in the soil. Meng and colleagues excluded all pathways other than the CMN by using barriers to keep the plants' root systems separate while allowing mycorrhizal strands through—a crucial step He and colleagues' study did not take.

Text 2

Meng and colleagues took the necessary precaution of separating the plants' root systems (thereby excluding root-to-root transmission). However, any barrier used must allow the thread-like hyphae of a CMN to pass through, and this permeability would also allow liquids through. Thus, the researchers' experimental design cannot ensure that any nutrient transfer observed can be attributed to a CMN and not to some other pathway.

8 Mark for Review

ABC

Based on the texts, how would the author of Text 2 most likely respond to the characterization of Meng and colleagues' study in Text 1?

A) By asserting that the author of Text 1 has overstated the effectiveness of the method that Meng and colleagues used

B) By pointing out that the author of Text 1 has overlooked studies that reported results that contradict those reported by Meng and colleagues

C) By claiming that the author of Text 1 has misrepresented what Meng and colleagues were trying to achieve with their experimental design

D) By arguing that the author of Text 1 has conflated the method used by Meng and colleagues with that used by He and colleagues

The fifteenth-century English Heege Manuscript is unusual among collections of its kind and time given its focus on fantasy tales over more acclaimed works by celebrated medieval authors like Hoccleve. But according to professor James Wade, even more unusually, the three texts in the manuscript's first booklet were likely copied by Richard Heege from a traveling minstrel's repertoire book. The evidence includes performative elements such as the narrator politely addressing the audience, joking about peasants and royalty, and making jokes that could be modified to refer to the town of Radford when in nearby Brackonwet to avoid giving offense.

9 Mark for Review ABC

As presented in the text, Wade would most likely agree with which statement about the first booklet of the Heege Manuscript?

A. It was likely a copy Heege intended to give to a traveling minstrel working in the area around Radford and Brackonwet.

B. The texts it includes were based on stories about the area around Radford and Brackonwet, but these names were later removed.

C. It was copied from a text that originated from a traveling minstrel who worked in the area around Radford and Brackonwet.

D. It was written down by Heege from memory based on a performance by a traveling minstrel who worked in the area around Radford and Brackonwet.

Within higher education, studying philosophy requires that students be conversant with the field's foundational texts and historical figures. By contrast, doing philosophy within or beyond the academy demands the creative, self- directed application of acquired expertise to enduring questions about the nature of existence and knowledge. While both approaches engage with influential figures, those who do philosophy treat such figures as vital interlocutors who facilitate new insights rather than as ossified authorities who, though relevant to the present, primarily represent the discipline's past.

10 Mark for Review

ABC

Based on the text, which choice best describes the relationship between doing philosophy and studying philosophy?

A. Doing philosophy involves developing novel ideas through imagined dialogue with past philosophers based on knowledge of those philosophers' views acquired by studying philosophy.

B. Doing philosophy helps students formulate concrete solutions to practical issues, whereas studying philosophy prioritizes engagement with historical arguments in the field.

C. Doing philosophy represents a departure from the norms that govern scholarly inquiry, whereas studying philosophy requires conformation to these norms.

D. Doing philosophy requires students to challenge the ideas articulated by past philosophers, especially when these ideas are broadly accepted by other people studying philosophy.

Researchers have identified over eighty gestures made by nonhuman great apes, such as clapping and bowing, that appear to convey information and that seem to be biologically inherited. Kirsty E. Graham and Catherine Hobaiter hypothesized that humans may be able to interpret great ape gestures, either through an evolutionary inheritance or as part of more general human cognitive abilities. The researchers tested this hypothesis by enlisting participants in an online game in which they had to correctly identify the meanings of ape gestures seen in videos. Though participants achieved some success, it is unclear whether they sometimes did so by making use of additional context provided by the images or sounds in the video recordings.

11 Mark for Review ABC

Which statement, if true, would most strongly support the underlined claim?

A) Participants were more readily able to identify an ape gesture when it meant "give me that food" than when it meant "let's be friendly."

B) Participants correctly identified gestures at the same rate for videos in which the apes made sounds in addition to gestures and videos in which the apes were silent.

C) Participants correctly interpreted ape gestures more than 50 percent of the time, whereas they would have only identified gestures correctly 25 percent of the time if they had been guessing.

D) When apes made mouth-touching gestures, which participants tended to correctly interpret as requests for food, the food was visible in the videos.

Throughout history, tall structures such as Ruwanwelisaya in Sri Lanka or the Minarets of Selimiye Mosque in Turkey have posed a challenge to their architects, since these structures are more susceptible to physical forces than smaller structures. For example, the lower floors of a building must be able to support the enormous weight of the upper floors. To test a hypothesis that steel is more suitable than concrete for construction of a tall building's lower floors, researchers created a computer model of two buildings that were identical except that one had concrete lower floors while one had steel lower floors, and simulated the effects of the forces of wind and gravity on those structures over the course of a century.

12 Mark for Review ABC

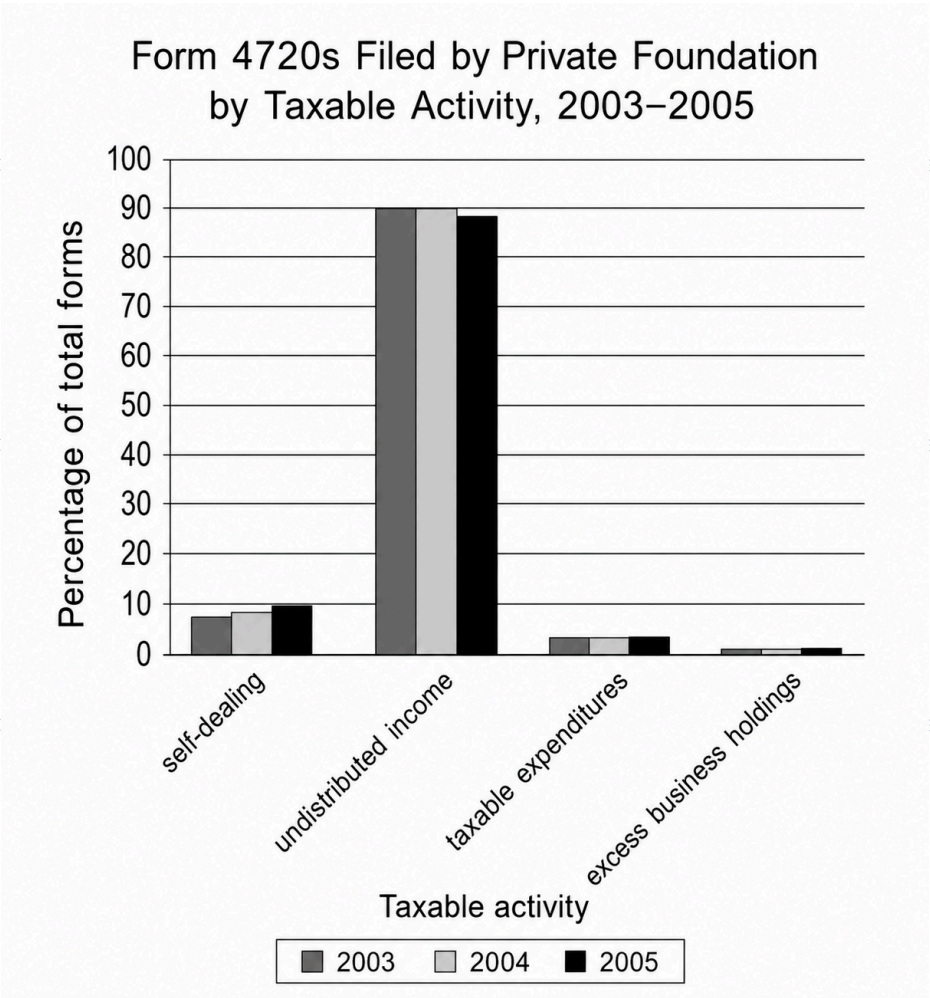
Which finding from the model tests, if true, would support the researchers' hypothesis?

A. At the end of the simulation, the structure with steel lower floors had slightly larger dimensions at its base than it did at the beginning of the simulation.

B. During the simulation, the structure with steel lower floors bent in high wind to a greater degree than did the structure with concrete lower floors.

C. During the simulation, when sudden forces caused vibrations in the structures, those vibrations took longer to disappear in the structure with steel lower floors than in the structure with concrete lower floors.

D. At the end of the simulation, the structure with concrete lower floors was several inches shorter than the structure with steel lower floors.



While US public charities, like Howard Hughes Medical Institute, must file Form 990 yearly with the IRS, private foundations, such as the W.K. Kellogg Foundation, must file a different form, 990-PF. In addition, foundations that engage in certain prohibited activities must also file Form 4720 and pay a penalty tax on the money involved. Private foundations are prohibited from holding excess interests in a business enterprise, "self-dealing" (conducting activities that benefit foundation insiders), making taxable expenditures such as outlays for lobbying, and failing to cross a required threshold in making charitable distributions from income. Out of the organizations that filed Form 990-PF in the years 2003–2005, _____

13 Mark for Review ABC

Which choice most effectively uses data from the graph to complete the assertion?

A) a greater percentage of those that also filed Form 4720 did so because they held excess interests in a business enterprise than the percentage of those that filed Form 4720 because they did not meet the minimum charitable distribution requirement.

B) more of those that also filed Form 4720 failed to meet a minimum charitable distribution requirement than did those organizations that filed Form 990 but also filed Form 4720 for the same reason.

C) those that were also required to file Form 4720 because they had excess holdings in a business enterprise paid, on average, a significantly smaller penalty than those organizations that filed Form 4720 because they engaged in self-dealing.

D) those that were also required to file Form 4720 were more likely to do so because they did not meet the minimum charitable distribution requirement than for other reasons.

Last-mile delivery refers to the final step in delivering packages to customers— and this final step can be very difficult. Delivery companies are contending with the increasing popularity of next-day delivery and with the increasing volume of parcels to be delivered, resulting in a growing bottleneck of packages in the last-mile delivery stage. These companies have been experimenting with innovative solutions in last-mile delivery, including self-service lockers (containers where consumers can pick up their parcels using a unique code). Unfortunately, many of these innovations create new obstacles (e.g., convenient placement of self-service lockers is challenging) and are not ready for full-scale implementation. As a result, delivery companies will likely _____

14 Mark for Review ABC

Which choice most logically completes the text?

A encourage consumers to expect next-day delivery only for products that are widely available.

B continue to struggle with last-mile delivery operations until viable solutions become available.

C be able to meet consumer's expectations for next-day delivery in many areas but not in all regions.

D have little incentive to find a solution to last-mile delivery challenges.

Across brown bears—omnivores with high dietary plasticity—there is wide variety in dietary mix, which may reflect genetics, local resource availability, or social learning (cubs stay with their mothers for two years or more). Evaluating these possibilities, Anne Hertel et al. analyzed 30 years of data on trophic position (indicative of dietary mix) for female brown bears. After separation, daughters, who tended to settle near their mothers, occupied the same trophic positions as their mothers for two years, but the correlation disappeared by year five. Trophic correlation with unrelated individuals in similar habitats was modest, while habitat-independent correlation with nonmaternal relatives (e.g., cousins) was no different than with unrelated individuals. These findings suggest that _____

15 Mark for Review ABC

Which choice most logically completes the text?

A) social learning and resource fluctuations may both play a role in dietary mix among females, at least temporarily, though genetic factors appear to make a significant contribution as well.

B) female dietary mix is best understood as changeable and contingent on fluctuating environmental conditions rather than as the result of social learning or genetic factors.

C) dietary mix among females may reflect a social learning effect that eventually diminishes, though environmental constraints cannot be ruled out as a contributing factor.

D) growing dissimilarity between mothers and their daughters with regard to dietary mix may reflect changes in the resources available in maternal habitats, though social learning could also contribute to the trend.

In economics, mustard seeds are considered a commodity because they are essentially interchangeable, "essentially" being a word that admits a degree of latitude. In 2021, researchers J. Shorish, M. Stephenson, and M. Zargham devised a mathematical model of fungibility (interchangeability) that would account for such variation. Distinguishing "exactly the same" and "nearly the same" commodities —whether crops, lumber, or precious metals— _____ among their primary aims.

16 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A having been

B being

C was

D were

Before the ratification of the 19th Amendment in 1920, many US territories extended voting rights to women, in part to spur westward migration that would help these territories meet population thresholds for statehood. The Territory of Wyoming approved women's sufrage in 1869 before attaining statehood in _____ while eastern states, lacking the same incentive, did not follow suit until after the turn of the century.

17 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A 1890, for example;

B 1890, for example,

C 1890; for example,

D 1890, for example:

Now seen as a mechanical precursor to the e-reader, Spanish educator and inventor Á ngela Ruiz Robles' s Enciclopedia Mecá nica eschewed pages in favor of three horizontal paper scrolls that could be easily swapped out to accommodate a range of subjects. Though Ruiz Robles's prototype had limited functionality, interactive features described in her 1949 patent— such as a button labeled "verb" that would illuminate relevant text when pressed— _____ an impressive early vision of hypertext.

18 Mark for Review ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A revealing

B reveal

C reveals

D has revealed

Traveling to India between 1935 and 1952, African American intellectuals Sue Bailey Thurman, Blanche Wright Nelson, and Sadie T.M. Alexander met with Mahatma Gandhi, an influential proponent of nonviolent resistance, and forged meaningful connections with local Indian women as part of an effort to promote women's _____ but also around the world.

19 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A empowerment— not only in India—

B empowerment not only in India,

C empowerment, not only in India,

D empowerment— not only in India

Jane Austen's Northanger Abbey (1818) is considered a satire of another novel popular at the time: Ann Radcliffe's The Mysteries of Udolpho (1794), which Austen's heroine, Catherine Morland, is depicted reading. However, the similarity of the _____ experiences— the predicaments of both Catherine and Radc life's Emily St. Aubert result from men's greed—suggests that underlying the satire is a social critique.

20 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A novel's protagonists'

B novels' protagonists

C novels' protagonists'

D novel's protagonist's

Roger Angell's reputation as one of the greatest baseball writers of all _____ bolstered by dozens of New Yorker essays in which he demonstrated a deep affection for and knowledge of the game, has perhaps overshadowed his remarkable career as a fiction editor: during his decades-long tenure at the New Yorker, Angell helped shape the work of John Updike, Ann Beattie, and other contributors of note.

21 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A time was

B time,

C time had been

D time, was

As part of his effort to identify Bronze Age pathways across Sicily, archaeologist Dario Calderone created three-dimensional digital models of the terrain via aerial photogrammetry. The drone _____ the landscape, which included archaeological sites from as far back as the Early Neolithic Period, flew in a pattern that ensured 60 percent forward overlap and 20 percent side overlap between individual images.

22 Mark for Review ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A photographed

B was photographing

C photographing

D photographs

Karel C pek’s 1920 play R. U. R. (Rossum 's Universal Robots), in which artificial workers overthrow their masters, left an indelible mark on the science fiction genre, and the English language, by introducing the term "robot" (derived from the Czech word robota, meaning "indentured labor" or "drudgery"). _____ C pek’s play also contributed to a venerable literary and mythological tradition: using artificial beings as mirrors and foils for humanity.

23 Mark for Review

ABC

Which choice completes the text with the most logical transition?

A Despite its creation of such an iconic trope,

B By achieving such a lofty goal,

C Ultimately limited in its lasting influence,

D Beyond the simple coining of a term,

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To determine the approximate age of stone tools excavated from ancient lake beds in Saudi Arabia, archaeologist Eleanor Scerri and colleagues collected samples of sediments surrounding the tools; these samples were then analyzed using a method known as optically stimulated luminescence (OSL) dating. _____ OSL dating indicated that the tools were around 400,000 years old.

24 Mark for Review

ABC

Which choice completes the text with the most logical transition?

A In addition,

B By comparison,

C Similarly,

D Ultimately,

Nineteenth-century Modernista architects championed nature in their designs. Granted, the wavy stone façade and ornate floral tilework of Casa Batlló, a Modernista private home designed by Antoni Gaudí, couldn't exactly grow in a forest. _____ one sees natural influences in Gaudí's penchant for curves (rather than right angles) and plant- and animal-inspired flourishes.

25 Mark for Review

ABC

Which choice completes the text with the most logical transition?

A Furthermore,

B Similarly,

C Still,

D In other words,

While researching a topic, a student has taken the following notes:

• Shanawdithit (1801–1829) was a Beothuk cartographer (mapmaker).
• Her maps of Newfoundland's Beothuk Lake outline both the lake and various points around the lake where encounters between the Indigenous Beothuk people and British colonists occurred.
• Her maps are notable for depicting the experiences the Beothuk had within the landscape.
• Contemporary Potawatomi cartographer Margaret Pearce: Indigenous cartography emphasizes "experienced space, or place, as opposed to the Western convention of depicting space as universal, homogenized, and devoid of human experience."
• Pearce: "Indigenous cartographies are as diverse as Indigenous cultures, from Hawaiian performative cartographies to Navajo verbal maps and sand paintings."

26 Mark for Review ABC

The student wants to describe Shanawdithit's approach and explain its significance. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Shanawdithit's maps are part of a broader tradition of Indigenous cartography that, according to Pearce, ranges from "Hawaiian performative cartographies to Navajo verbal maps and sand paintings."

B) Shanawdithit mapped Beothuk Lake through significant encounters that occurred there, an approach that Pearce describes as "depicting space as universal [and] homogenized."

C) By depicting experiences of the Beothuk that occurred around Beothuk Lake, Shanawdithit's maps reflect Indigenous cartography's emphasis on "experienced space, or place" rather than the landscape alone.

D) According to Pearce, Indigenous cartography, such as Shanawdithit's maps of Beothuk Lake, emphasizes "experienced space, or place," with a variety of approaches that reflect the diversity of Indigenous cultures.

While researching a topic, a student has taken the following notes:

- It is generally accepted that all bats use echolocation to navigate.
- The statement "all bats use echolocation to navigate" is falsifiable.
- A falsifiable statement can theoretically be proved false by observation.
- "All bats use echolocation to navigate" could be falsified by observing a single bat that didn’t use echolocation.
- The statement "some bats use echolocation to navigate" is not falsifiable.
- Falsifying it would require observing every bat in the universe, which is not possible.

27 Mark for Review

ABC

The student wants to contrast the falsifiability of two statements about bats. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. One could theoretically observe a single bat that didn’t use echolocation; however, one could not falsify the statement "some bats use echolocation to navigate. "

B. The statement "some bats use echolocation to navigate" could be falsified by observing every bat in the universe, but it is not possible to falsify the statement "all bats use echolocation to navigate."

C. The statement "all bats use echolocation to navigate" could be proved false by observing a single bat that didn’t use echolocation, whereas falsifying "some bats use echolocation to navigate" is impossible.

D. While it is generally accepted that all bats use echolocation to navigate, the statement "some bats use echolocation to navigate" is not falsifiable.

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Reading & Writing

Module 2

27 Questions

In an analysis of over 50,000 records of pollinators visiting flowers, botanists determined that flowers with bilateral symmetry (zygomorphs) are visited by a smaller subset of pollinators than the subset that visits flowers with radial symmetry (actinomorphs), thus _____ anecdotal observations that zygomorphic flowers exclude many potential pollinators.

1 Mark for Review

ABC

Which choice completes the text with the most logical and precise word or phrase?

A allocating

B substantiating

C transposing

D supplanting

Rare tree species growing in the Tyson Research Center are much likelier than common species to be found in conspecific (same-species) aggregations, but in the Barro Colorado Island plot, conspecific aggregation and abundance are nearly _____, which is to be expected given that rare species in Barro Colorado are heavily reliant on seed dispersion via animals, while those in the Tyson Center tend to drop seeds directly to the ground.

2 Mark for Review ABC

Which choice completes the text with the most logical and precise word or phrase?

A inverted

B indistinguishable

C independent

D intermittent

The preponderance of studies on aqueous uranium (U)—including a 1997 study by Moran et al.—were in oceanic contexts. In a 2020 investigation of factors influencing the abundance of the isotope U-238 in Kaluka Lake and three other lakes in northwest China, Zhao et al. therefore aimed to address the relative _____ of scientific knowledge on uranium isotopes in inland bodies of water.

3 Mark for Review

ABC

Which choice completes the text with the most logical and precise word or phrase?

A dearth

B aggregation

C diminution

D obsolescence

The early twentieth-century modernist movement is often characterized in terms of revolutionary artists' struggle against a static middle-class culture; by contending that the latter was itself _____, literary scholar Michael Levenson frames the conflict between the two not as a clash between dynamism and stagnation but instead as "a contest of novelties."

4 Mark for Review ABC

Which choice completes the text with the most logical and precise word or phrase?

A attenuated

B yielding

C evolving

D entrenched

Visiting flowers that produce high volumes of nectar allows hummingbirds, including stripe-throated hermits, to maximize their foraging efficiency. But nectar can't be seen from outside the flower, and while some studies have suggested that petal length may indicate nectar volume, María A. Maglianesi and team found no such association among hummingbird-visited plants at Los Cusingos Bird Sanctuary and other sites in Costa Rica. The researchers hypothesize that by driving hummingbirds to visit more plants in search of high-reward flowers, plants' lack of reliable signaling may increase chances of pollination.

5 Mark for Review ABC

Which choice best describes the function of the underlined portion in the text as a whole?

A. It attributes the inconclusive findings reported in the text regarding the influence of a particular floral trait on interactions between hummingbirds and plants to differences in study sites.

B. It suggests that a hypothesis discussed later in the text regarding a potential visual cue that hummingbirds rely on to increase their foraging efficiency has little support from the broader scientific community.

C. It acknowledges research findings, which weren't replicated by the study discussed later in the text, positing a relationship between floral traits that potentially underlies hummingbird foraging decisions.

D. It summarizes contradictory study findings concerning a floral trait that influences hummingbird foraging patterns, which the study mentioned later in the text aimed to resolve.

A trend away from stable, relatively high humidity in the Middle East over the past 8 million years can be inferred from the composition of cave formations dating to different periods. Cave formations from about 7 million years ago mainly consist of transparent columnar calcite, indicative of an underground water system regularly replenished by rainfall. Cave formations younger than approximately 1 million years, however, mainly consist of branching, opaque (and sometimes colorful) material, often with more frequent growth interruptions, indicating an intermittent water supply.

6 Mark for Review ABC

Which choice best describes the function of the two underlined portions in the text as a whole?

A) They identify different characteristics of cave formations that when considered together provide evidence for a long-term shift toward lower and more variable humidity in the Middle East.

B) They describe divergent features of cave formations that when taken together imply decreasing variation in Middle Eastern rainfall over time.

C) They draw a contrast between Middle Eastern cave formations that is suggestive of geographic differences in the underground water supply.

D) They provide evidence from different Middle Eastern cave formations that substantiates the claim that the region experienced a long period of relatively high humidity.

In many parts of the world, merchants rarely label goods with exact prices like $17.31 or $203.62. Studies have shown that consumers view exact prices as reflecting merchant expertise and careful calculation, so why aren't they more common? One reason is that exact prices may create greater cognitive friction during purchase decisions than do whole-number prices or prices easily rounded to whole numbers (e.g., $5.99, which is little different from $6). Particularly if multiple goods are being considered for purchase, tracking and comparing exact prices may feel onerous to consumers.

7 Mark for Review

ABC

Which choice best states the main purpose of the text?

A. To argue that exact prices are relatively uncommon because calculating them is burdensome for merchants

B. To address the question of why the use of exact prices is less common in some parts of the world than in others

C. To explain why exact prices present greater cognitive friction for some consumers than whole-number prices or easily rounded prices do

D. To account for the fact that exact prices remain uncommon in many places despite some positive consumer perceptions of them

Text 1

Coal mining and other human activities are significant sources of atmospheric methane (CH_4), but, as Sander Houweling and colleagues have shown, so too are naturally occurring emissions, such as those from wild animals. This biogenic CH_4 is challenging to monitor given natural sources' variability and spatial diffusion, but one promising approach is to use CH_4 flux data from extant monitoring infrastructure as proxies for CH_4 fluxes in unmonitored areas.

Text 2

Sparkle Malone and colleagues examined the United States' network of eddy covariance (EC) towers—the equipment used to track CH_4 fluxes—to evaluate the utility of network data for quantifying biogenic CH_4. Malone et al. found that EC towers are disproportionately concentrated in areas where biogenic CH_4 emissions are known to be elevated, a pattern that seriously compromises the representativeness of network data.

8 Mark for Review ABC

Based on the texts, Malone et al. (Text 2) would most likely agree with which statement about the "approach" described in Text 1?

A) It could compromise the accuracy of the biogenic CH_4 emissions data that EC towers collect in areas with elevated emissions.

B) It would only be viable if the EC towers monitoring biogenic CH_4 emissions could detect smaller fluxes than they can at present.

C) It may suggest that biogenic CH_4 emissions in unmonitored areas are higher than they actually are.

D) It is likely to be useful for determining biogenic CH_4 emissions in unmonitored areas of some regions but not of most regions.

Maltese lithic (stone implement) assemblages from the sixth and seventh millennia BCE exhibit notable characteristics. Nearly all implements were fashioned from limestone—mostly from beach cobbles—rather than superior chert or obsidian, and the assemblages predominantly comprise flakes produced through direct percussion, with negligible representation of cores, blades, or retouched tools. The fairly rudimentary nature of Malta's lithic industry parallels contemporaneous Sardinian technologies rather than the sophisticated geometric microlith traditions documented in neighboring Sicily.

9 Mark for Review

ABC

Which choice best states the main idea of the text?

A. Although Malta, Sardinia, and Sicily all show evidence of lithic industry in the sixth and seventh millennia BCE, the characteristics of the stone tools in each location suggest three distinct levels of technological sophistication.

B. Maltese stone tools from the sixth and seventh millennia BCE are notable for their relative simplicity and their similarity with stone tools produced in Sicily.

C. While not definitive evidence, the materials and methods used in the production of Maltese stone tools in the sixth and seventh millennia BCE are strongly suggestive of the exchange of lithic technology between Malta, Sardinia, and Sicily.

D. The characteristics of Maltese stone tools from the sixth and seventh millennia BCE are indicative of a level of lithic technological development that, while not uniquely low, was less advanced than that observed nearby in Sicily.

The following text is adapted from Richard Connell's 1923 short story "The Man Who Could Imitate a Bee." An ornithologist is a scientist who studies birds.

At twelve [years old Hervey Deyo] had resolved to be a scientist and furthermore a great scientist. He determined to pursue the career of ornithologist; there was something so dignified and withal scientific about a science that called the sparrow *Passer domesticus* and the robin *Erithacus rubecula*. He made rapid progress. On his thirteenth birthday he took a bird walk at dawn and was able to record in his note-book the scientific names of forty-nine birds, including the ruby-and-topaz humming-bird (*Chrysolampis mosquitus*) which is rare around Boston.

10 Mark for Review

ABC

According to the text, what motivates Hervey Deyo to become an ornithologist rather than another kind of scientist?

A His early success in identifying rare birds

B His general ambition to become a great scientist in any field

C The scientific names used for birds

D The opportunity to take peaceful morning walks in nature

A 2025 study performed by Simon Townsend and colleagues presents evidence that bonobos, close relatives of chimpanzees, are capable of a rudimentary compositionality in communication—that is, they can organize sounds they make such that the order carries meaning in addition to the sounds themselves. After reviewing bonobo calls recorded in forests of the Democratic Republic of the Congo, Townsend's team claimed that bonobos sometimes paired calls in specific orders and that the contexts of such pairs indicate they were deliberate communications rather than products of chance. The team based their claim on mathematical analysis of hundreds of hours of bonobo calls along with notes of the contexts in which they occurred.

11 Mark for Review ABC

Which finding, if true, would best support the researchers' claim?

A. A bonobo grunt seems to function as a request for other bonobos to join in whatever activity is ongoing, since bonobos emitting a grunt did so during potentially collaborative contexts such as feeding, nesting, and grooming.

B. Bonobos used a single yelp or a single peep as an imperative, but a yelp appeared to be closer to a command, while a peep tended to function as a suggestion.

C. When bonobos paired grunts with other calls, they did not use rules of syntax as humans do to combine groups of sounds into even larger units of meaning.

D. While a single whistle from a bonobo was often an attempt to organize groups, a whistle preceded by a peep sound usually occurred in the presence of a charged social situation.

A student is writing a research paper on the history of irrigation in the state of California, situating the development of the Crystal Springs Reservoir (created in San Mateo County, California, in 1888) in a larger historical context. The student claims that innovation in irrigation techniques was driven in part by the climate in some parts of the state.

12 Mark for Review

ABC

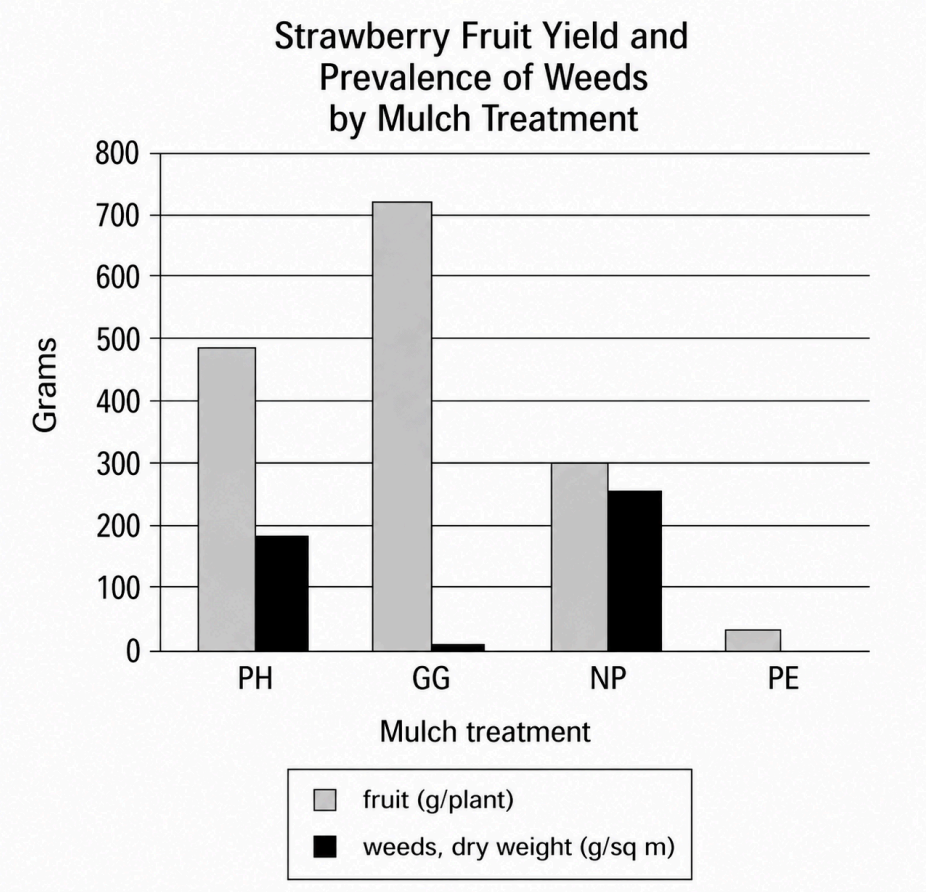
Which quotation from a study of California agriculture best supports the student's claim?

A "The necessity of irrigation infrastructure in California today cannot be overstated, since it is the most common means of conveying water for agricultural use."

B "The irrigation system developed by the Hohokam people in the 7th century CE in what is now Arizona was simple but applied hydraulic engineering design features that are still used today in both Arizona and California."

C "While the low humidity of Southern California's desert areas makes irrigation a requirement if agriculture is to be a success there, the relatively high humidity in the northern parts of the state makes such practices less necessary."

D "Competition for scarce water resources in Southern California inspired the development of sprinkler irrigation systems, a manner of irrigating that requires less use of water than traditional surface irrigation methods."



Polyethylene (PE) plastic sheeting is used as agricultural mulch because it is effective at reducing the propagation of weeds. Seeking a biodegradable alternative, Waqas Ahmad and colleagues tended otherwise equivalent plots of strawberries mulched with PE sheeting or one of several newsprint pulp–based alternatives: just newsprint pulp (NP), pulp bound with guar gum (GG), or pulp bound with psyllium husk (PH). The graph represents strawberry plots for which all emerging weeds were left in place throughout the experiment. Once the fruit was harvested, the weeds were removed, dried, and weighed for dry biomass. Based on the results, there was not a predictable association between treatment effectiveness and fruit yield.

13 Mark for Review ABC

Which choice most effectively uses data from the graph to support the underlined claim?

A. The psyllium husk and guar gum treatments were associated with lower weed biomasses and higher fruit yields than the treatment with just newsprint pulp was.

B. While the guar gum and polyethylene treatments were associated with the lowest weed biomasses, the guar gum treatment was associated with the highest fruit yield and the polyethylene treatment was associated with the lowest fruit yield.

C. The treatment with just newsprint pulp was associated with the highest weed biomass and was not one of the two treatments associated with the highest fruit yields.

D. While the psyllium husk treatment was associated with a higher weed biomass than the guar gum treatment was, the guar gum treatment was associated with a higher fruit yield than the psyllium husk treatment was.

At Cayo Largo (Puerto Rico), coral colonies experiencing degradation—a problem facing reefs globally—have been enhanced with transplanted fragments of elkhorn (a stony coral). This is a useful restoration approach but is challenging at scale since it involves cultivating and individually moving young stony coral pieces. In 2025, Samapti Kundu and her team announced a new approach: coating reef substrates with a simple-to-apply substance, made with inexpensive and abundant core components (e.g., gelatin), that can mimic a signal that draws coral larvae to settle in healthy reefs. When tested in conditions simulating real ocean environments, the substance enhanced settlement by up to twentyfold. These attributes suggest that the substance _____

14 Mark for Review ABC

Which choice most logically completes the text?

A offers a means of reef restoration that likely allows much broader implementation than the approach used at Cayo Largo does.

B should be able to simulate healthy reef signals even more closely when it is applied in natural ocean conditions.

C is likely to require more costly components to ensure its stability when used for coral colonies as degraded as those at Cayo Largo.

D should be most beneficial for restoring reefs that predominantly consist of elkhorn coral colonies.

Whereas Rebecca M. Harrison and Pia Nystrom's 2001 study of captive bonobos reported more right-handedness than left-handedness, Jane Goodall's 1963 study of wild chimpanzees did not. According to a meta-analysis of studies of nonhuman primates, captive populations are more likely to be described as right-handed than wild populations are. Statistical analysis indicates a handedness study would need a minimum of 176 individuals to show a representative result; however, the study by Harrison and Nystrom included a total population of 19, and the study by Goodall included a total population of 8. This suggests that _____

15 Mark for Review ABC

Which choice most logically completes the text?

A Goodall likely underestimated the prevalence of right-handedness among the wild chimpanzees in the study.

B the study by Harrison and Nystrom reliably represents handedness in captive primates but not in wild primates.

C the study by Harrison and Nystrom reliably represents handedness in captive primates, but the study by Goodall likely does not reliably represent handedness in wild primates.

D neither the study by Harrison and Nystrom nor the study by Goodall provides sufficient evidence to make a meaningful comparison about handedness in primates.

The Nimatron was an early digital computer designed to play a mathematical game against visitors to the 1939–1940 New York World's Fair. Though capable of making game moves in milliseconds, this one-ton machine—built by Westinghouse Electric Corporation engineers to demonstrate how electrical relays could respond to players' choices according to predetermined mathematical patterns—_____ programmed with time delays to give the impression of deliberative thinking.

16 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A were

B was

C being

D are

While pointe shoes, worn by ballerinas so they can balance on their toes, originated as simple satin slippers, some now incorporate modern materials or shock-absorption techniques inspired by athletic footwear. Such innovations, developed to address the increase in physically demanding choreography in modern ballet, _____ the evolution of the pointe shoe that began centuries ago.

17 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A has continued

B continue

C continues

D to continue

California is one of several US states with legislative provisions for the incorporation of worker cooperatives, businesses in which each worker receives an equal vote in company governance. Skeptics have questioned the co-op model for decades and declared _____ means of governance economically unsustainable, ignoring many examples of successful worker-owned businesses.

18 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A its

B they’re

C their

D one’s

Researchers recently theorized that Copper Age (3500 BCE to 2750 BCE) owl-shaped carvings collected from Iberian archaeological sites were likely toys created by children. This hypothesis, based on comparative analyses of owl drawings made by modern children, is supported by the objects predominantly being crafted from slate—a material that, because of its _____ be easily carved by small hands using rudimentary tools.

19 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A malleability could—

B malleability, could

C malleability could

D malleability—could

When Indhu Rubasingham was announced as the National Theatre's new artistic director in late 2023, theater leadership across the UK _____ increasingly corporate in structure, with a growing number of venues experimenting with models that would place executives in leadership, eliminating the artistic director position entirely.

20 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A became

B was becoming

C has become

D would become

In 2022, 73 percent of Mexico's exports, in terms of dollar value, were manufactured _____ Mexico relies on manufactured goods for the majority of its exports places it in a more stable economic position, generally speaking, than countries that rely primarily on raw materials, which are more sensitive to market fluctuations.

21 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A goods, that

B goods that,

C goods. That

D goods that

Many writing systems used in South and Southeast Asia use the same punctuation mark to indicate textual breaks. Each of these scripts, which include Devanagari and other Brahmic systems, _____ a punctuation mark called the danda to indicate subdivisions in text, with single and double dandas alternated depending on the length of the preceding text.

22 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

A are deploying

B deploying

C deploys

D deploy

In the Pakistani truck art tradition, elaborately decorated cargo vehicles feature poetic verses, religious messages, or slogans that reflect drivers' values, aspirations, or humor. _____ they invest considerably in decorating their trucks, seeing them both as personal artistic creations and as contributions to their country's folk art tradition.

23 Mark for Review ABC

Which choice completes the text with the most logical transition?

A Indeed, owners prize their vehicles as canvases for creative expression:

B Finally, owners complete the artistic process after months of planning:

C However, owners must balance these aesthetic elements with practical functionality:

D As a result, owners expect significant financial returns from their decorated vehicles:

In a 2022 analysis of small sovereign countries, Archie Simpson categorized forty-one as microstates based, in part, on the criterion of having a population under one million—for instance, San Marino, with its population of approximately 34,000. _____ political scholar Sarina Theys suggests an intersubjective approach to classification by which a small country's status is self-defined rather than relying on externally imposed or quantitative standards.

24 Mark for Review

ABC

Which choice completes the text with the most logical transition?

A Challenging the accuracy of San Marino's population data,

B In addition to this example,

C Reinforcing the importance of Simpson's population criterion,

D Countering this method of categorization,

Researchers have long been aware that pistol shrimp use a specialized claw to create bubbles that stun their prey, but understanding of the phenomenon has been limited by researchers' inability to accurately measure the tiny, rapid movements of the shrimp. _____ scientists have turned to computer simulations to re-create the claw's closing motion, allowing them to track the pressure, speed, and density of water throughout the bubble formation.

25 Mark for Review

ABC

Which choice completes the text with the most logical transition?

A Addressing the flaws in their hypothesis,

B In light of this limitation,

C To determine where they went wrong,

D To correct this misconception,

While researching a topic, a student has taken the following notes:

• Sarah Hinckley et al.'s 2009 study of planktonic ecosystems used a nutrient-phytoplankton-zooplankton (NPZ) model.
• Diane Stoecker et al. (2017): NPZ models rely "on the differentiation of the planktonic communities into phytoplankton or zooplankton functional types," excluding mixotrophic plankton.
• Phytoplankton are exclusively phototrophic, producing nutrients via photosynthesis.
• Zooplankton are exclusively phagotrophic, obtaining nutrients by consuming other organisms.
• Mixotrophic plankton are capable of both phototrophy and phagotrophy.
• Stoecker et al.: "Mixotrophy changes the basic trophic structure of…ecosystems; most of the unicellular organisms at the base of the plankton food web can no longer be regarded as strictly producers or consumers."

26 Mark for Review ABC

The student wants to challenge the model used in Hinckley et al.'s study. Which choice most effectively uses information from the notes to accomplish this goal?

A. Since it classifies plankton as exclusively phototrophic producers or phagotrophic consumers, the model used in Hinckley et al.'s study may misrepresent planktonic ecosystems, in which—as Stoecker et al. note—many plankton combine these functions.

B. Hinckley et al.'s 2009 study used an NPZ model, which, as Stoecker et al. point out, unnecessarily categorizes organisms as either phytoplankton, zooplankton, or mixotrophic plankton.

C. By restricting plankton functional types to phytoplankton and zooplankton, the model used in Hinckley et al.'s study of planktonic ecosystems overlooks the many mixotrophic species that—as Stoecker et al. note—function as neither producers nor consumers.

D. Hinckley et al.'s model disregards the respective roles of producers and consumers—organisms that, as Stoecker et al. emphasize, are essential to the trophic structure of the plankton food web.

While researching a topic, a student has taken the following notes:

• Ynés Mexía was a Mexican American botanist.

• Between 1917 and 1938, she collected over 150,000 botanical samples throughout the Americas.

• She collected a sample of *Mutisia speciosa* in Minas Gerais, Brazil, in 1930.

• She collected a sample of *Baccharis salicifolia* in Nayarit, Mexico, in 1926.

• *Mutisia speciosa* and *Baccharis salicifolia* are part of the Asteraceae family.

• These specimens can now be viewed online at the C.V. Starr Virtual Herbarium.

27 Mark for Review ABC

Which choice most effectively uses information from the given sentences to introduce Mexía to a new audience?

A Between 1917 and 1938, Ynés Mexía collected samples of the plants *Mutisia speciosa* and *Baccharis salicifolia*, among others.

B Botanist Ynés Mexía is known for collecting over 150,000 botanical samples throughout the Americas between 1917 and 1938.

C Many of the botanical samples collected by Ynés Mexía, such as *Mutisia speciosa* and *Baccharis salicifolia*, can now be viewed on the C.V. Starr Virtual Herbarium.

D Among the many botanical samples Ynés Mexía collected in Brazil in 1930 was *Mutisia speciosa* of the Asteraceae family.

Answer Key

Reading & Writing — August Predictions (Hard Mode)

Module 1

1 **B**	2 **D**	3 **A**	4 **B**	5 **D**	6 **A**	7 **A**	8 **A**	9 **C**
10 **A**	11 **D**	12 **D**	13 **D**	14 **B**	15 **C**	16 **C**	17 **B**	18 **B**
19 **D**	20 **C**	21 **B**	22 **C**	23 **D**	24 **D**	25 **C**	26 **C**	27 **C**

Module 2

1 **B**	2 **C**	3 **A**	4 **C**	5 **C**	6 **A**	7 **D**	8 **C**	9 **D**
10 **C**	11 **D**	12 **D**	13 **B**	14 **A**	15 **D**	16 **B**	17 **B**	18 **A**
19 **B**	20 **B**	21 **C**	22 **C**	23 **A**	24 **D**	25 **B**	26 **A**	27 **B**

MATH

Module 1

22 Questions

@isatprep

1

The functions p and q are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $p(x) = 25(1.2)(2)^{x+2}$

II. $q(x) = 120(2)^{x}$

A. I only

B. II only

C. I and II

D. Neither I nor II

2

Consider the polynomial below.

$$9x^2 + 48xy + 64y^2$$

Which of the following expressions is a factor of the polynomial above?

A) 9x + 64y

B) 3x + 64y

C) 3x + 8y

D) 9x + 8y

3

$$V(t) = 25{,}000(0.88)^t$$

A construction company purchased a new heavy-duty loader. The function above models the estimated value of the loader, in dollars, t years after it was purchased in 2018, where $t \leq 12$. Which of the following is the best interpretation of 25,000 in this context?

A. The estimated value of the loader, in dollars, in 2018

B. The estimated value of the loader, in dollars, 12 years after 2018

C. The estimated percent decrease in the loader's value each year after 2018

D. The estimated percent decrease in the loader's value every 12 years after 2018

4

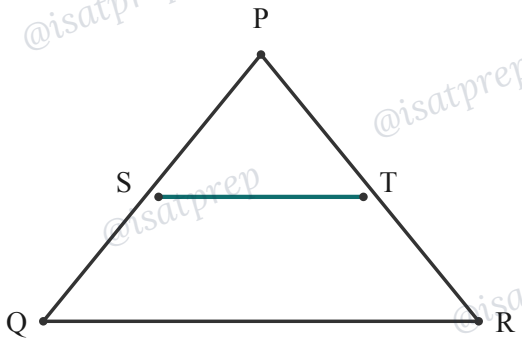
$$16.4x + 13.15y = 575$$

A coffee shop owner purchased pounds of Ethiopian coffee and pounds of Sumatran coffee, which cost a total of $575. The given equation represents the relationship between the number of pounds of Ethiopian coffee, x, and the number of pounds of Sumatran coffee, y, purchased. How much more, in dollars, did a pound of Ethiopian coffee cost than a pound of Sumatran coffee?

ANSWER

5

In the figure shown, triangle PQR is similar to triangle PST. The measure of angle PST is 42°, and $QR = 12(ST)$. What is the measure of angle PQR?



A $(12 \cdot 42)°$

B $(12 + 42)°$

C 42°

D 12°

6

$$12x + 8 = k(6x + 8) + 6x$$

In the given equation, k is a constant. The equation has exactly one solution. Which value **CANNOT** be the value of k?

A -1

B 0

C 1

D 2

7

Line j is defined by $4x + 5y = 25$. Line k is parallel to line j in the xy-plane. An equation of line k is $16x + ry = 15$, where r is a constant. If line k passes through the point $(0, b)$, what is the value of b?

ANSWER

8

$$f(x) = 21(1.20)^{x/6}$$

For the given function f, the value of $f(x)$ increases by p% for every increase of x by 6. What is the value of p?

ANSWER

9

A model estimates that the initial number of photons in an X-ray beam is 500 when the beam reaches the surface of a certain material. The model also estimates that when the beam passes through this material, the number of photons in the beam decreases by 50% for each 11 millimeters of the material the beam passes through. Which equation represents this model, where I is the estimated number of photons in the beam when the beam reaches a point x millimeters from the surface?

A $I = 500(0.5)^{x/11}$

B $I = 500(0.5)^{x}$

C $I = 500(2)^{x/11}$

D $I = 500(11)^{x/2}$

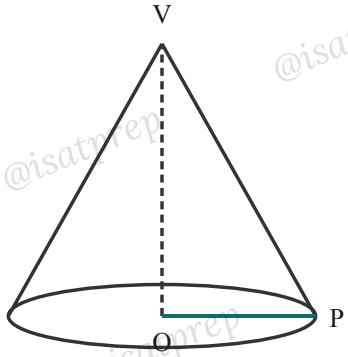
10

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 2900% of the number of prey at the start of the week, and the number of predators had increased by 320% of the number of predators at the start of the week. The number of predators at the end of the week was p% less than the number of prey at the end of the week. What is the value of p?

ANSWER

11

For the right circular cone shown, OP is a radius of the base and V is the vertex. The cone's lateral surface area, in square meters, is $rL\pi$, where r is the length, in meters, of OP and L is the length, in meters, of VP. The cone has a base area of 81π square meters and a lateral surface area of 214π square meters. What is the height of the cone, to the nearest meter?



ANSWER

12

If n and k are numbers greater than 1 and $\sqrt[n]{(n^5)}$ is equivalent to $\sqrt[3]{(k^2)}$, for what value of a is n^{2a+1} equal to k?

ANSWER

13

A computer program models the population of a certain insect in an environment where the insect has no natural predators. According to the model, the estimated total mass of the population of insects at the end of every 5-week period is 158% greater than the estimated total mass at the end of the previous 5-week period. The estimated total mass of the population of insects at the end of 15 weeks is 627.7 grams. Which equation best represents this model, where M is the estimated total mass, in grams, of the population of insects at the end of t weeks?

A $M = 538.92(1.58)^{t/5}$

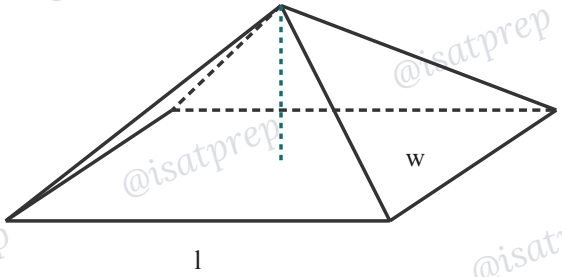
B $M = 457.66(2.58)^{t/5}$

C $M = 159.14(1.58)^{t/5}$

D $M = 36.55(2.58)^{t/5}$

14

The figure shown is a right rectangular pyramid, where l = 12 units, w = 6 units, and h = 14 units. What is the surface area, in square units, of the pyramid?



ANSWER

15

In triangle JKL, the measure of angle J is $90b°$, the measure of angle K is $66a°$, and the measure of angle L is $24a°$, where a and b are constants. Which of the following must be true?

A. $\cos L > \sin K$

B. $\cos L = \sin K$

C. $\cos L < \sin K$

D. There is not enough information to compare the values of cos L and sin K.

16

$$(2/7)x^2 - 6x + \sqrt{(7k + 4)}x + \sqrt{(7k + 4)} = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 49. What is the value of k?

ANSWER

17

The expression $6x^4 + 17x^2 + 5$ can be rewritten as $(3x^2 + a)(2x^2 + b)$ where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$ where c and d are positive non-integers. What is the value of $a + c$?

ANSWER

18

The height of a right circular cylinder is 59 inches, and the circumference of its base is 590 inches. Which expression represents the total surface area, in square inches, of the cylinder?

A $59{,}590$

B $\pi(295^2/\pi) \cdot 59$

C $59{,}590 + \pi(295/\pi)^2$

D $59{,}590 + 2\pi(295/\pi)^2$

19

$$9x^2 + 8 = nx$$

In the given equation, n is a constant. The equation has exactly one solution. What is the value of $n^2/8$?

ANSWER

20

An isosceles right triangle has a perimeter of $34 + 34\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

A 17

B $17\sqrt{2}$

C 34

D $34\sqrt{2}$

21

$$h(x) = -\sqrt{(x^2 + bx + c)}$$

The function h is defined by the equation above, where b and c are constants. In the xy-plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{358})$. If $h(m) = 0$, what is the greatest possible value of m?

ANSWER

22

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$, and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy-plane, which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the corresponding function?

I. $f(x) = a(4.2)^{x+b}$

II. $g(x) = a(4.2)^{x} + b$

A. I only

B. II only

C. I and II

D. Neither I nor II

MATH

Module 2

22 Questions

@isatprep

1

Which expression is a factor of $81p^{21} - 121p^{19}$?

A $9p + 11$

B $81p^2 + 121$

C $11p - 9$

D $-40p^{44}$

2

The exponential function g is defined by $g(x) = ab^x$, where a and b are positive constants. If $g(c) = m$ and the value of $g(c + 1)$ is 75% more than the value of m, where c and m are constants, what is the value of b?

A 0.25

B 0.75

C 1.75

D 3.5

3

$$44x^2 + (44b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab, where k is a constant. What is the value of k?

A $\frac{1}{44}$

B $\frac{1}{11}$

C 1

D 44

4

Cylinder A and cylinder B are each right circular cylinders with a radius of 4 inches. The height of cylinder A is half the height of cylinder B, and the total surface area of cylinder A is 128π square inches. If the cylinders are glued together along a circular base, what is the height, in inches, of the resulting cylinder?

A) 8

B) 12

C) 24

D) 36

5

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(11) > 0$, $f(14) < 0$, and $f(17) > 0$, what is one possible value of $a + b$?

27 [ ] 29

6

$$x - a = (x - a)(x - 27)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 28$?

I. a
II. 27
III. 28

A) I and II only

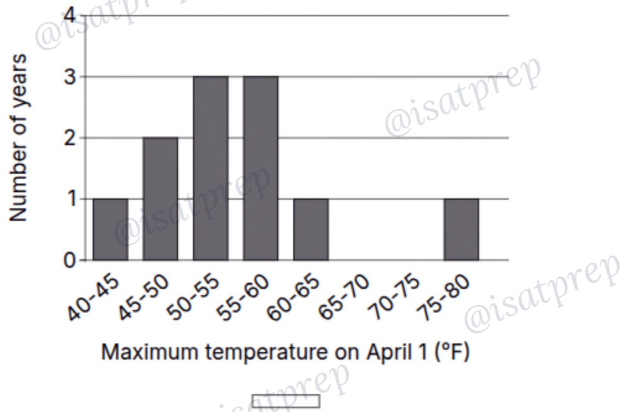
B) I and III only

C) II and III only

D) I, II, and III

7

each year at a certain weather station for 11 years. The histogram summarizes the recorded data set.



The temperature of 77.4 degrees Fahrenheit is removed from this data set to create a new data set of 10 temperatures. Which of the following statements must be true?

I. The mean of the new data set is less than the mean of the original data set.
II. The median of the new data set is less than the median of the original data set.

A I only

B II only

C I and II

D Neither I nor II

8

An online store is offering two different discounts on its sweaters: a bulk discount for large orders and a coupon discount. For orders of n sweaters, where $10 < n < 50$, a bulk discount of $n\%$ off the original price of 39 dollars is applied to the price of each sweater. If a coupon is applied to the order, a discount of an additional 3 dollars is applied to the price of each sweater, after the bulk discount is applied. For a certain order with both the bulk discount and the coupon discount applied, the total price after the discounts for the sweaters purchased was 729 dollars. How many sweaters were purchased in this order?

A) 19

B) 20

C) 30

D) 44

9

$$34x^2 + (34b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab, where k is a constant. What is the value of k?

A $\frac{1}{34}$

B $\frac{1}{17}$

C 1

D 34

10

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(25) > 0$, $f(28) < 0$, and $f(31) > 0$, what is one possible value of $a + b$?

55 | | 57

11

An online store is offering two different discounts on its sweaters: a bulk discount for large orders and a coupon discount. For orders of n sweaters, where $10 < n < 50$, a bulk discount of $n\%$ off the original price of 32 dollars is applied to the price of each sweater. If a coupon is applied to the order, a discount of an additional 2 dollars is applied to the price of each sweater, after the bulk discount is applied. For a certain order with both the bulk discount and the coupon discount applied, the total price after the discounts for the sweaters purchased was 612 dollars. How many sweaters were purchased in this order?

A) 19

B) 20

C) 30

D) 46

12

In triangle XYZ, the measure of angle X is 90 degrees. Point W lies on segment YZ, and segment WX is perpendicular to segment YZ. The length of segment WY is 548, and the length of segment WX is 411. What is the value of $\tan Z$?

A $\frac{4}{3}$

B $\frac{4}{5}$

C $\frac{3}{4}$

D $\frac{3}{5}$

13

In triangle ABC, the measure of angle B is 90 degrees and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 23 and the length of $\overline{AC}$ is 11 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

A $\frac{11}{23}$

B $\frac{23}{34}$

C $\frac{34}{23}$

D $\frac{23}{11}$

14

$$x - a = (x - a)(x - 35)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 36$?

I. a
II. 35
III. 36

A) I and II only

B) I and III only

C) II and III only

D) I, II, and III

15

A right square pyramid has a surface area of $100 + 20\sqrt{554}$ square inches, which includes a base area of 100 square inches. What is the height, in inches, of this pyramid?

16

$$2x^2 - x\sqrt{k} + 2y^2 + y\sqrt{t} - 12 = 0$$

The given equation, where k and t are positive constants, defines a circle in the xy-plane. The radius of this circle is $\sqrt{57}$. Which expression represents the value of t?

A $180 - k$

B $k - 720$

C $720 - k$

D $816 - k$

17

$$x^2 + (\sqrt{k-3})x + 26 = 0$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k?

A 29

B 101

C 104

D 107

18

$$f(x) = 9((\frac{1}{8})^x + 4)$$

The function f is defined by the given equation. For all values of x, the value of $f(x)$ is greater than k, where k is a constant. What is the greatest possible value of k?

A) 0

B) 4

C) 36

D) 288

19

x	y
-5	$91 + 4k$
5	91
10	$91 - 2k$

The table gives three values of x and their corresponding values of y, where k is a positive constant. There is a linear relationship between x and y. Which equation represents this relationship?

A $2kx - 5y = 91 + 2k$

B $2kx - 5y = 455 + 10k$

C $2kx + 5y = 91 + 2k$

D $2kx + 5y = 455 + 10k$

20

If $x = \sqrt[3]{2x^n + 75}$, where n is a positive integer constant, what is the value of x^n?

21

Which expression is a factor of $x^4 + 14ax^2 + 49a^2 - 64$, where a is a positive constant?

A $x^2 + 7a + 8$

B $x^2 - 7a - 8$

C $x^2 + 14a$

D $x^2 - 8a$

22

The estimated total pressure a diver experiences below the surface of the water is equal to the sum of the estimated water pressure and the estimated atmospheric pressure. At a depth of h meters below the surface, the estimated total pressure P_1, in kilopascals (kPa), a diver experiences is given by $P_1 = 10h + 101$. After maintaining a depth of h meters, the diver descended d additional meters. The estimated total pressure P_2, in kPa, the diver experienced after descending these d additional meters is given by $P_2 = 10d + 341$. What is the estimated total pressure, in kPa, the diver experienced at a depth of h meters below the surface?

A) 442

B) 341

C) 101

D) 10

ANSWER KEY

MATH — MODULES 1 & 2

MODULE 1

1. B
2. C
3. A
4. 3.25
5. C
6. C
7. 3/4
8. 20
9. A
10. 86
11. 22
12. (15-2n)/(4n)
13. D
14. 335.20
15. B
16. 192/7
17. 8.5
18. D
19. 36
20. B
21. 179
22. D

MODULE 2

1. A
2. C
3. A
4. D
5. 28
6. B
7. A
8. C
9. A
10. 56
11. C
12. A
13. C
14. B
15. 23
16. D
17. D
18. C
19. D
20. 25
21. A
22. B